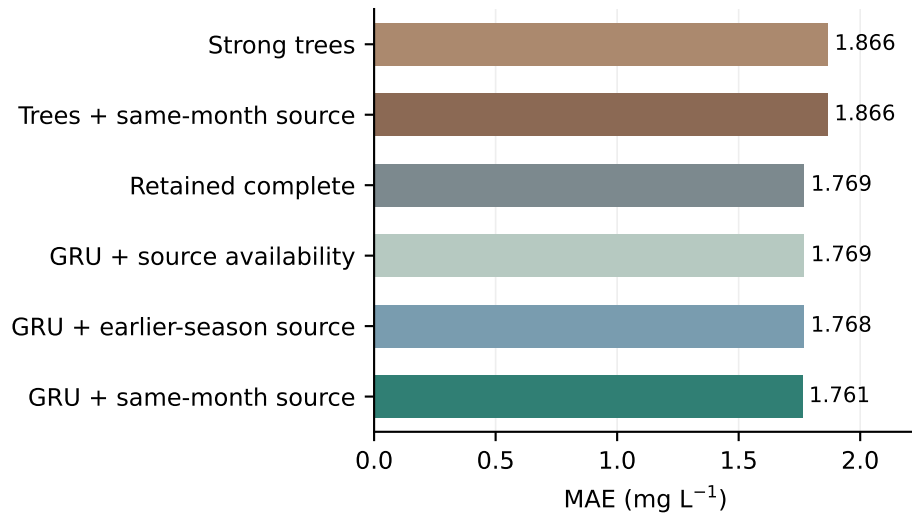


# Learning time-aligned source DOC information in the existing GRU

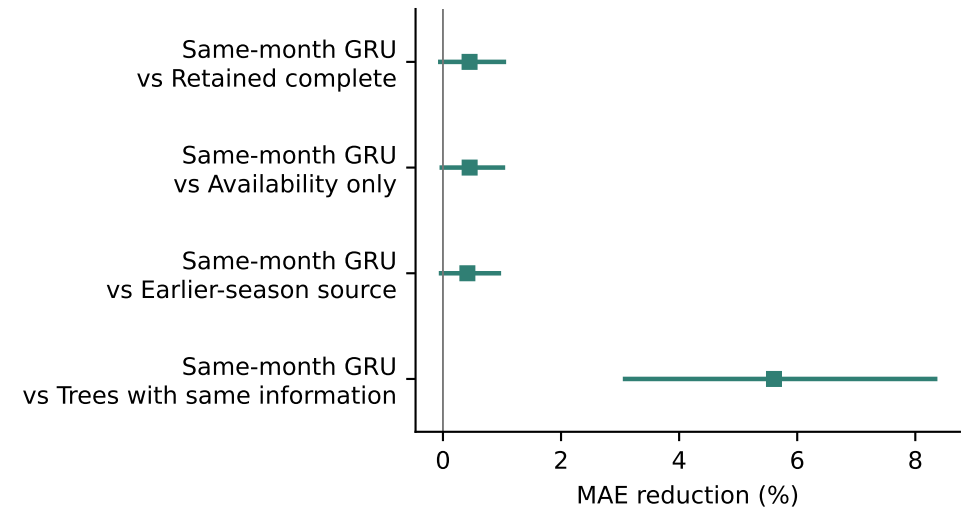
**a**

## Unmonitored-site DOC reconstruction



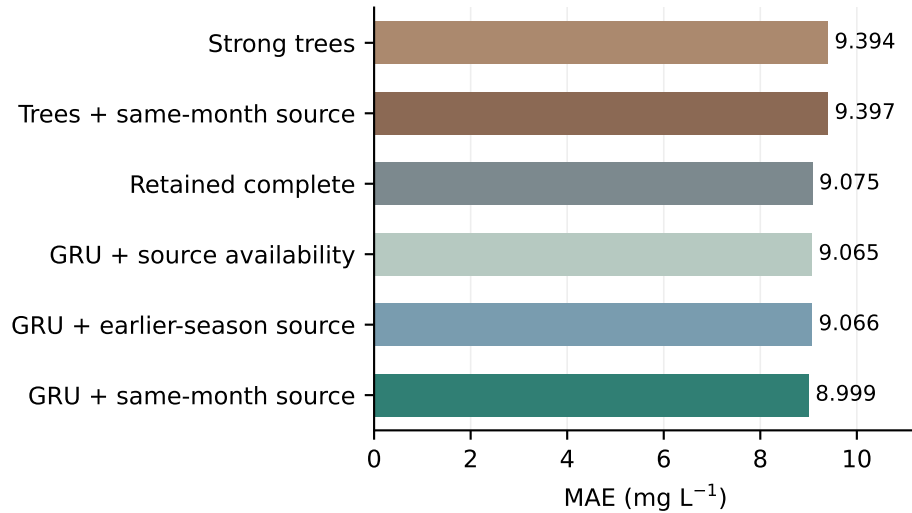
**b**

## 95% paired station intervals



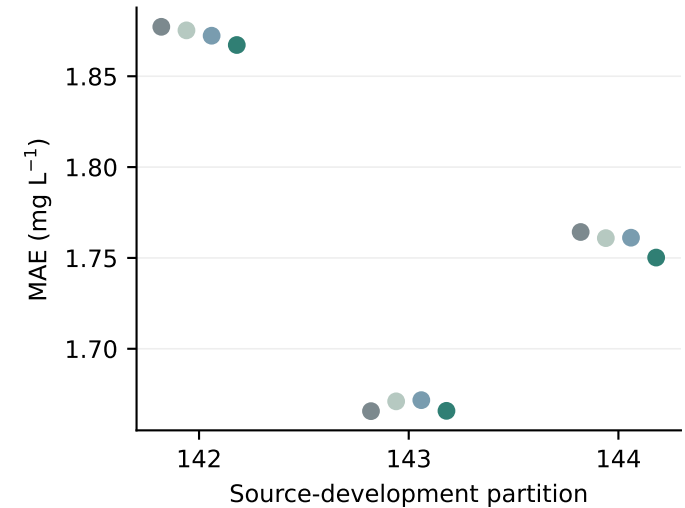
**c**

## High-DOC reconstruction



**d**

## Matched partition averages



● Retained complete
 ● GRU + earlier-season source
 ● GRU + source availability
 ● GRU + same-month source

3 partitions × 3 seeds · Same reference/backbone · Matched support and causal historical controls  
 Receiving water quality absent · 5,000 station draws · Source-development selection; geographical/external results unchanged